

Hardware Inventory - Functional Requirements Document

Functional Requirements Document (FRD)

Product: Hardware Inventory System

Document type: Functional Requirements

Audience: Shop owners, managers, and stakeholders (also usable by developers for acceptance)

Version: 1.0

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Related docs: [FSD.md](#) (functional design), [DATA_DICTIONARY.md](#) (schema), [ER_DIAGRAMS.md](#) (ER), [USER_GUIDE.md](#) (how to operate), [packaging/GO-LIVE-NOTES.md](#)

1. Document control

Item	Value
Scope basis	As-built application (code + schema + go-live notes)
Companion design	Functional Specification Document (docs/FSD.md)
Requirement ID pattern	FR-<AREA>-nn (functional), NFR-<AREA>-nn (non-functional)
Priority	Must = required for intended shop operation; Should = expected when feature is enabled; Could = optional / flagged off by default

This FRD states **what** the system must do and **who** it serves. Detailed flows, APIs, and rules are in the FSD.

2. Purpose and product summary

The Hardware Inventory System is a **retail point-of-sale (POS) and inventory console** for a hardware / electrical / paints shop. It supports:

- Product catalog with units, barcodes, GST-related fields, and pricing
- Purchasing from suppliers and stock adjustments
- Counter sales with cash / online / split payments and credit (outstanding) balances
- Invoice reprint and on-demand business reports
- Optional **scheduled owner reports** delivered by email and saved in-app

Currency and tax context are **India-oriented** (INR, GST/HSN fields, Indian financial-year quarters for scheduled reports, Asia/Kolkata scheduling).

Brand display name is configurable at build time (e.g. shop name); default branding is “Raj Hardware.” Invoice letterhead is set separately.

3. Stakeholders and actors

Actor	Description
Shop owner / Admin	Full back-office: catalog, purchases, adjustments, reporting, schedule config, price review
Manager	Same effective access as Admin (legacy role treated as admin)
Cashier	Counter work: POS, outstanding payments, invoice reprint; view-only products and stock overview
System scheduler	In-process job that generates scheduled reports on a cron (IST)
Email recipient	Owner email that receives “report ready” links when SMTP is configured

4. Assumptions and constraints

ID	Assumption / constraint
A-01	Single shop deployment; one PostgreSQL database
A-02	Staff authenticate with username + password; browser holds a JWT (no server-side session revocation list)
A-03	Deployment may be local Windows/Linux or hosted (e.g. Railway); UI may be served by the API or a separate Vite origin
A-04	Production requires a strong JWT_SECRET and configured DATABASE_URL
A-05	Scheduled report periods and cron use Asia/Kolkata ; quarterly periods follow the Indian financial year (Apr–Mar)
A-06	Bulk CSV/Excel imports are capped at approximately 1,500 rows per file

ID	Assumption / constraint
A-07	Promotions and several inventory sub-features may be hidden by frontend feature flags

5. In scope and out of scope

In scope

- Login, session, and admin “acting as” another user
- Home dashboard shortcuts, activity, and admin-oriented revenue views
- Product catalog CRUD (admin), view (cashier), bulk import, price review
- Suppliers (admin) and customers (all authenticated users for create/list as implemented)
- Purchases with stock-in and supplier payments (admin)
- Stock overview and manual stock adjustments (adjustments admin-only)
- POS sales, payments, outstanding balances, invoice reprint
- On-demand reporting (admin, when reporting flag is on)
- Scheduled owner reports: configuration, generation, history, optional email
- Appearance settings (light / dark / system) in the browser
- Document number series for bill vs tax invoices (env-configurable start values)
- Health check and production packaging/backup practices documented separately

Out of scope (current product)

- Native mobile apps
- Multi-branch / multi-company tenancy
- Online customer storefront or e-commerce
- Full GST return filing / e-invoice government integration
- Server-side theme or company profile settings API (invoice letterhead is build-time / code config)
- Features whose UI flags are off by default and not part of the primary go-live path (e.g. promotions UI, unused inventory sub-navs) — documented only as optional when enabled

6. Functional requirements

Priority legend: **Must** | **Should** | **Could**

6.1 Authentication and session

ID	Requirement	Priority	Acceptance criteria
FR-AUTH-01	Users shall sign in with username and password and receive	Must	Valid credentials return token + user profile; invalid credentials return unauthorized without issuing a token.

ID	Requirement	Priority	Acceptance criteria
	an access token for subsequent API calls.		
FR-AUTH-02	Protected API operations shall require a valid Bearer token for an active user.	Must	Missing/invalid/expired token yields 401; inactive users cannot authenticate.
FR-AUTH-03	Failed login attempts shall be rate-limited per client IP.	Must	After the configured max failures in the window, further attempts are rejected until the window resets (configurable; default 10 / 15 minutes).
FR-AUTH-04	Admin/Manager users shall be able to switch the effective (“acting”) user when multiple staff accounts exist.	Should	Cashiers cannot switch; admins can select another active user; permissions follow the acting user where enforced.
FR-AUTH-05	The client shall restore session state after reload while the token remains valid.	Must	Reloading the app with a stored valid token returns the user to an authenticated console without re-entering credentials.

6.2 Home / dashboard

ID	Requirement	Priority	Acceptance criteria
FR-HOME-01	After login, users shall land on a Home view with navigation to primary tasks.	Must	Home shows shortcuts appropriate to role.
FR-HOME-02	Home shall surface recent activity and stock-related alert counts from live catalog data.	Should	Low/out-of-stock style counts reflect current product stock/reorder data.
FR-HOME-03	Admin/Manager Home shall include revenue / timeseries oriented dashboard data when reports are available.	Should	Admin sees revenue chart or equivalent; cashier focuses on sell shortcuts.

6.3 Catalog (products)

ID	Requirement	Priority	Acceptance criteria
FR-CAT-	Admin/Manager shall create, update, and	Must	Mutations succeed for admin; cashiers cannot mutate via UI/API.

ID	Requirement	Priority	Acceptance criteria
01	delete products with SKU, name, units, pricing, and GST-related fields (HSN, tax %).		
FR-CAT-02	Cashiers shall view the product catalog in read-only mode.	Must	Cashiers can open Products and see data but cannot add/edit/delete.
FR-CAT-03	Products shall support multi-unit definitions and optional barcodes.	Must	Units convert to a base unit; barcodes uniquely identify products/packs as configured.
FR-CAT-04	Brand codes shall be unique when present (case-insensitive).	Must	Duplicate brand codes are rejected; blank brand code may auto-generate (e.g. BC-#####).
FR-CAT-05	Admin shall import products from CSV/Excel up to ~1,500 rows per batch.	Should	Template available; successful import creates products and shows a concise success summary.
FR-CAT-06	Large catalogs shall be browsable with server-side pagination and search.	Must	Products table loads a page size (e.g. 50) with search, not the entire catalog at once in the grid.
FR-CAT-07	When a purchase lowers cost, the system shall keep current selling price and flag the product for price review with a suggested price.	Must	Flagged products appear for admin review; apply/dismiss works per item and in bulk.
FR-CAT-08	When purchase cost rises, selling price may increase according to markup rules without requiring a review flag.	Must	Documented go-live behavior: cost up → shelf price can rise via markup; cost down → review flag.

6.4 Suppliers and customers

ID	Requirement	Priority	Acceptance criteria
FR-PARTY-01	Admin/Manager shall maintain suppliers (name, contact, GST, etc.).	Must	Supplier list/create available to admin; used on purchases.
FR-PARTY-	Authenticated users shall be able to list	Must	Customer records can be linked or snapshot on sales.

ID	Requirement	Priority	Acceptance criteria
02	and create customers with optional GST/state fields for invoicing.		

6.5 Purchases and supplier payments

ID	Requirement	Priority	Acceptance criteria
FR-PUR-01	Admin/Manager shall record purchases from a supplier with line items (product, quantity, unit cost).	Must	Recording a purchase increases stock and updates cost/pricing per business rules.
FR-PUR-02	Admin shall import purchase lines from CSV/Excel (up to ~1,500 lines) matching catalog by brand code and/or name.	Should	Required columns validated; large imports show summary UI; Record Purchase commits stock.
FR-PUR-03	Admin shall record payments against purchases and track purchase balance outstanding.	Should	Payments reduce balance; unpaid received purchases appear in supplier outstanding reporting.
FR-PUR-04	Cashiers shall not access purchase create/list APIs or the Purchases nav.	Must	Non-admin receives forbidden / nav hidden.

6.6 Point of sale (POS)

ID	Requirement	Priority	Acceptance criteria
FR-POS-01	Authenticated users shall create sales by searching products, setting quantities/units, and completing payment.	Must	Completed sale decreases stock and assigns a sale number.
FR-POS-02	Checkout shall support Cash and Online banking payment methods.	Must	Method is stored on payment records for reporting.

ID	Requirement	Priority	Acceptance criteria
FR-POS-03	Checkout shall support split cash + online amounts; unpaid remainder remains balance due.	Must	Split validation prevents over-allocation; remainder is outstanding, not auto-filled.
FR-POS-04	Sales shall support discounts, optional transport/freight amount, and walk-in or registered customer details (including tax invoice party fields where used).	Should	Totals reflect lines + discount + tax + transport; party GST/state captured when provided.
FR-POS-05	Concurrent sales shall not oversell the same product stock.	Must	One sale succeeds; conflicting sale returns insufficient stock.
FR-POS-06	Bill vs tax invoice numbering shall use separate configurable series.	Should	Numbers follow configured starts (e.g. BIL-n / INV-n).

6.7 Outstanding balances

ID	Requirement	Priority	Acceptance criteria
FR-OUT-01	Users shall list completed sales with remaining balance due.	Must	Outstanding list shows unpaid/partial sales.
FR-OUT-02	Users shall record additional payments against an outstanding sale.	Must	Payment reduces balance; method recorded.

6.8 Invoices (reprint)

ID	Requirement	Priority	Acceptance criteria
FR-INV-01	Users shall search or look up sales by number and reprint/view invoice documents.	Must	Reprint flow retrieves sale by number/id and renders invoice for print.

6.9 Stock overview and adjustments

ID	Requirement	Priority	Acceptance criteria
FR-STK-01	Authenticated users shall view inventory overview with stock levels and status (in stock / low / out).	Must	Overview searchable/sortable as implemented.
FR-STK-02	Admin/Manager shall post stock adjustments with reason (and optional note), updating on-hand quantity with an audit trail.	Must	Before/after quantities stored; stock movements recorded; cashiers cannot adjust.

6.10 On-demand reporting

ID	Requirement	Priority	Acceptance criteria
FR-RPT-01	When reporting is enabled, Admin/Manager shall run date-range reports for sales summary, sales by product/customer, purchases, supplier payments, gross margin, and tax invoice sales.	Must	Each report returns data for the selected from/to range for admin only.
FR-RPT-02	Admin shall access dashboard timeseries / revenue series for charts.	Should	Home/reporting charts load for admin.
FR-RPT-03	Cashiers shall not access reporting APIs or the Reporting nav.	Must	Forbidden / nav hidden.

6.11 Scheduled owner reports

ID	Requirement	Priority	Acceptance criteria
FR-SCH-01	Admin shall configure which report types are enabled, their period (daily/monthly/quarterly/yearly), and a notify email.	Must	Config persists; email required if any type enabled.
FR-SCH-	The system shall generate scheduled reports for the	Must	Runs appear in history with status and payload when

ID	Requirement	Priority	Acceptance criteria
02	previous completed period for: Sales, Inventory snapshot, Supplier outstanding, Customer outstanding.		successful.
FR-SCH-03	When SMTP and notify email are configured, the system shall email a link to open the saved report in the app.	Should	Without SMTP, runs still save; email is skipped with operational awareness.
FR-SCH-04	Scheduler shall run in Asia/Kolkata on a configurable cron, staggering report types to avoid overload.	Should	Default morning stagger; one type processed per tick; idempotent completed runs.
FR-SCH-05	Admin shall open a saved report run from history or deep link (?reportRun=).	Must	Viewer shows the stored report content for authorized admin.

6.12 Promotions (optional)

ID	Requirement	Priority	Acceptance criteria
FR-PROMO-01	When the promotions feature flag is enabled, Admin shall create/update/delete percentage promotions (cart / product / category scope).	Could	UI and mutate APIs available only when flag on; cashiers cannot mutate.

6.13 Settings and branding

ID	Requirement	Priority	Acceptance criteria
FR-SET-01	Users shall choose Light, Dark, or System appearance preference stored in the browser.	Should	Preference persists locally; no server round-trip required.
FR-SET-02	Company display name used in the UI shall be configurable at frontend build time.	Should	Changing Vite company env vars updates visible brand strings after rebuild.

6.14 Quotations

ID	Requirement	Priority	Acceptance criteria
FR-QUO-01	Admin/Manager shall create quotations for an existing customer or for a new customer who is not yet in the customer master.	Must	A quotation can be issued with customerId set or null. New-customer details are not required to exist in Customer first. Optional “Save as customer” creates or links a customer by phone without blocking issue.
FR-QUO-02	Quotation lines shall reference existing active products and use Product.sellingPrice. The client cannot submit a unit price or totals.	Must	Missing selling price or an inactive product is rejected. Stored unitPrice matches selling price × unit conversion. No second product catalog is created.
FR-QUO-03	Creating, editing, or issuing a quotation shall not change stock, sales, purchases, or payments.	Must	Product.currentStock and stock movements are unchanged. No Sale row is created.
FR-QUO-04	The server shall number quotations QUO-YYYY-#### and set validity to 2 calendar days from the quotation date in Asia/Kolkata.	Must	Numbers are unique under concurrent creates. validUntil is the quotation date plus 2 days. Dates are not taken from the browser.
FR-QUO-05	Discount, GST, transport, and totals shall follow the tax-invoice calculation and be recomputed on the server with decimal arithmetic.	Must	Line and order discounts, CGST/SGST/IGST from the product rates, taxable amount, and grand total match that invoice math. Client totals are rejected.
FR-QUO-06	Draft quotations can be edited. Issued quotations cannot be edited. Cancelled quotations cannot be issued. Expired is derived when the current India date is after validUntil.	Must	Status transitions above are enforced. Historical quotations are not deleted.
FR-QUO-07	Admin/Manager shall list, view, print, and download a customer-facing quotation PDF.	Must	List filters by number/customer, status, and date. PDF shows business letterhead, quotation number, dates, customer snapshot, lines, and totals.

7. Role permission matrix

Effective admin = ADMIN or MANAGER.

Capability	Admin / Manager	Cashier
Login / JWT session	Yes	Yes
Switch acting user	Yes	No
Home, POS, Outstanding, Invoices	Yes	Yes
Quotations	Yes	No
Products — view	Yes	Yes
Products — create/update/delete/import/price review	Yes	No
Inventory overview	Yes	Yes
Purchases & suppliers	Yes	No
Stock adjustments	Yes	No
Reporting (on-demand)	Yes	No
Scheduled owner reports	Yes	No
Promotions mutate (if flag on)	Yes	No
Settings (theme)	Yes	Yes
Customers list/create	Yes	Yes

8. Non-functional requirements

ID	Requirement	Priority	Acceptance criteria
NFR-SEC-01	Production shall require a strong JWT secret and restrict browser CORS origins as configured.	Must	Misconfigured production JWT fails closed; CORS allows listed origins (+ same-origin when applicable).
NFR-SEC-02	Passwords shall be stored hashed (not plaintext).	Must	Seed/login uses password hashes.
NFR-CON-01	Stock updates for concurrent sales shall be concurrency-safe (no oversell).	Must	See FR-POS-05.
NFR-CON-02	Quotation numbers shall be allocated inside a transaction with a year-row lock so concurrent creates cannot collide.	Must	See FR-QUO-04.
NFR-PERF-01	API shall accept large JSON bodies sufficient for ~1,500-	Must	Bulk import does not fail with 413 under default limit.

ID	Requirement	Priority	Acceptance criteria
	row imports (default 15 MB).		
NFR-PERF-02	Bulk product create and large purchase record shall complete without routine transaction timeouts for supported batch sizes.	Should	Chunked/batched writes as implemented.
NFR-OPS-01	Operators shall be able to back up and restore the PostgreSQL database using provided scripts.	Must	Documented backup/restore path works on a maintained installation.
NFR-OPS-02	GET /api/health shall report API liveness without authentication.	Must	Returns OK-style JSON when process is up.
NFR-OBS-01	API requests shall be logged with method, path, status, and duration.	Should	Visible in server stdout for operations.
NFR-I18N-01	Money presentation and invoice wording target INR / Indian retail practice.	Should	UI and invoices use rupee formatting conventions as implemented.

9. Traceability (FR → FSD)

FR area	FSD section
FR-AUTH-*	§4 Auth & RBAC
FR-HOME-*	§6.1 Home
FR-CAT-*	§6.2 Catalog
FR-PARTY-*	§6.3 Parties
FR-PUR-*	§6.4 Purchases
FR-POS-*	§6.5 POS
FR-OUT-*	§6.6 Outstanding
FR-INV-*	§6.7 Invoices
FR-STK-*	§6.8 Stock
FR-RPT-*	§6.9 Reporting
FR-SCH-*	§6.10 Scheduled reports
FR-PROMO-*	§6.11 Promotions

FR area	FSD section
FR-SET-*	§6.12 Settings
FR-QUO-*	§6.8a Quotations
NFR-*	§7 Cross-cutting & §9 Deployment

10. Glossary

Term	Meaning
Acting user	Effective user identity for permissions and audit fields after optional admin switch
Base unit	Canonical stock unit for a product; other units convert via factors
Outstanding	Remaining unpaid balance on a completed sale (or unpaid purchase balance for suppliers)
Price review	Flag when cheaper purchase cost suggests a lower shelf price pending owner decision
Report run	Saved scheduled report instance with period bounds and JSON payload

End of FRD. Behavioral design is specified in docs/FSD.md.